

Cubic Tornheim Derivatives and Harmonic Stieltjes Constants

A two-coordinate formula, a digamma-cube bridge,
and finite reductions at mixed integer orders

Research continuation for the ProveIt project

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Abstract

This report develops three connected exact-identity results. First, a coordinate-normalized integral of the digamma cube is expressed through the first strict harmonic Stieltjes coefficient, linking an ordered Hurwitz Laurent coordinate to the third derivative of diagonal Tornheim zeta. A normally convergent harmonic series and an explicitly normalized antiderivative supply independent analytic descriptions. Second, every admissible third directional derivative of Tornheim zeta at its singular origin is reduced to two fixed constants and ordinary Riemann-zeta derivatives. The coefficient of each nonclassical coordinate is explicit; an asymmetric conic of directions eliminates one coordinate. A transverse curved version retains these two coordinates and has an exact fourth-jet correction. Third, products containing arbitrary positive and nonpositive integer polylogarithm orders admit finite nested-polylogarithm reductions with a free complex outer order. The construction includes integer weights, repeated colors, all outer-order derivatives, and the logarithmic Mellin poles that distinguish the mixed sector from the purely rational one. Proofs, exact symbolic certificates, independent numerical checks, source-status notes, and further research questions accompany the results. No arithmetic independence of the displayed constants is asserted.

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Contents

1	Research scope and conventions	3
1.1	What is proved here	3
1.2	Classical input and the extent of novelty	3
1.3	Notation	3
2	The cubic digamma finite part and the first harmonic coefficient	4
2.1	A global partial-fraction identity	5
2.2	Reduction to one logarithmic harmonic series	6
2.3	A normalized complex antiderivative	7
2.4	A short certified evaluation route	8
3	The diagonal Tornheim derivative in harmonic coordinates	8
3.1	The completed cubic generating function	8
3.2	An explicit reduction of the third diagonal derivative	9
4	The complete cubic directional layer of Tornheim zeta	10
4.1	Statement of the formula	11
4.2	A holomorphic remainder with exactly two visible polar divisors	11
4.3	Exact slices determine the constant and linear remainder	12
4.4	Proof of the cubic formula	13
4.5	An independent convergent integral for the second coordinate	14
4.6	Coordinate cancellation and exact examples	14
5	Transverse curved directions and a sharp fourth-jet correction	15
5.1	A correction involving only the lower directional layer	16
6	Arbitrary depth with mixed integer inner orders	16
6.1	Definitions and finite input data	17
6.2	The mixed-sign reduction theorem	18
6.3	Continuation, poles, and normal differentiation	20
6.4	Explicit mixed and weighted identities	22
6.5	A compatible primitive hierarchy and exact checks	23
7	Verification, source corrections, and further research	24
7.1	What the checks establish	24
7.2	A source-status ledger for integration	25
7.3	Corrections and qualifications worth preserving	25
7.4	Proposed further research questions	26
	References	28

1 Research scope and conventions

The current ProveIt manuscript already contains extensive exact polylogarithm, Gamma, Stieltjes, and correlation identities. Its newest incoming reports go further: they construct arbitrary-depth ordered Hurwitz germs, complete coincident Stieltjes products, and compute the first two derivatives of Tornheim zeta on every admissible ray through the origin [1, 3, 2]. The questions addressed here are specific continuations of that work.

1.1 What is proved here

1. The cubic digamma finite part is reduced to the strict harmonic coefficient $\eta(1)$ already used in the ordered Hurwitz report:

$$\text{FP} \int_0^1 \psi(x)^3 dx = 3\zeta(2) - 6\eta(1) - 6\gamma\gamma_1.$$

This identifies two previously separate coordinates in the inspected reports. It supplies a convergent harmonic-series answer to the proposed reduction of the diagonal Tornheim third derivative. It does not give a finite evaluation of $\eta(1)$ using only ordinary zeta constants.

2. The entire cubic directional layer of $\mathcal{T}(A, B, C)$ is determined by two fixed constants. An exact double-zeta product identity reduces the number of unknown quadratic coefficients in its holomorphic remainder; the resulting formula answers the asymmetric-ray question in Section 9, Question 2 of the coincident report.
3. The mixed integer-order problem in Section 9, Question 11 of that report has a finite constructive solution at arbitrary depth. The free outer spectral order is retained throughout. The pole description includes logarithmic singularities, which are absent from the purely rational case.

The Gaussian S_6 and revised S_8 candidates are different arithmetic questions. They are not proved by the results below. The exact-identity and source-status ledgers retain this distinction.

1.2 Classical input and the extent of novelty

The Mellin representation, Jonquière expansion, shuffle and stuffle products, Mittag-Leffler methods, and the elementary relationship between Hurwitz zeta and polygamma functions are classical [8, 9, 10, 7, 6]. General Laurent algorithms for multiple zeta functions also predate this report [5]. Harmonic Stieltjes coefficients have an established normalization and literature [4].

The contribution is the specific cubic bridge, the complete formula for arbitrary cubic directions and transverse curves, and the explicit finite reduction and pole prescription for the mixed integer-order sector. These are proved directly. They are new continuations relative to the inspected repository corpus; a claim of global priority for every equivalent formulation would require a broader literature audit. In particular, a two-coordinate spanning formula is not a proof that the corresponding numbers are arithmetically independent.

1.3 Notation

The generalized Stieltjes convention is

$$\zeta(1+u, a) = \frac{1}{u} + \sum_{m=0}^{\infty} \frac{(-1)^m \gamma_m(a)}{m!} u^m, \quad \gamma_m = \gamma_m(1), \quad \gamma = \gamma_0. \quad (1.1)$$

Thus $\gamma_0(a) = -\psi(a)$ and $\psi = \Gamma'/\Gamma$. Primes on $\zeta(s, a)$ denote derivatives in its first argument unless explicitly stated otherwise. We set

$$L = \log(2\pi), \quad H_n^{(r)} = \sum_{j=1}^n j^{-r}, \quad H_n = H_n^{(1)}, \quad H_0^{(r)} = 0.$$

Powers of positive real numbers use real logarithms. All logarithms of complex variables in a right half-plane use the branch real on the positive axis.

Our ordered double zeta convention places the larger summation index first:

$$Z_2(s, t; a) = \sum_{n>m\geq 0} (n+a)^{-s}(m+a)^{-t}, \quad a > 0. \quad (1.2)$$

The series converges, for example, if $\Re s > 1$ and $\Re(s+t) > 2$. Its meromorphic continuation is understood when a spectral argument is outside this domain. The strict harmonic coefficient is fixed by

$$Z_2(1+u, 1; a) = \frac{1}{u^2} + \frac{\gamma_0(a)}{u} + \frac{\gamma_0(a)^2 - \zeta(2, a)}{2} + \eta(a)u + O(u^2). \quad (1.3)$$

This is the convention of the ordered incoming report. If $\zeta_H(s, a)$ uses the inclusive inner sum, then

$$\zeta_H(s, a) = Z_2(s, 1; a) + \zeta(s+1, a), \quad \eta(a) = -\gamma_H(1, a) - \zeta'(2, a). \quad (1.4)$$

The correction $\zeta'(2, a)$ and the minus sign are essential.

For a function with an endpoint expansion in powers of x and $\log x$, $\text{FP} \int_0^1$ means the constant coefficient of the cutoff integral $\int_\varepsilon^1$ as $\varepsilon \downarrow 0$, in the coordinate x itself. We never replace this prescription silently by a spectral Laurent constant or by a nonlinear cutoff coordinate.

2 The cubic digamma finite part and the first harmonic coefficient

In this section write $\eta = \eta(1)$ and $Z_2(s, 1) = Z_2(s, 1; 1)$, with the conventions fixed in Section 1. In particular, $B_2 = (\gamma^2 - \zeta(2))/2$ and

$$Z_2(1+u, 1) = u^{-2} + \gamma u^{-1} + B_2 + \eta u + O(u^2).$$

The inclusive harmonic coefficient is related by $\eta = -\gamma_H(1, 1) - \zeta'(2)$ [4, 3].

Theorem 2.1 (Cubic-harmonic bridge). *In the unit-coordinate Hadamard finite-part convention,*

$$Q := \text{FP} \int_0^1 \psi(x)^3 dx = 3\zeta(2) - 6\eta - 6\gamma\gamma_1, \quad (2.1)$$

$$= 3\zeta(2) + 6\gamma_H(1, 1) + 6\zeta'(2) - 6\gamma\gamma_1. \quad (2.2)$$

An absolutely convergent arithmetic representation is

$$Q = 3\zeta(2) + 6\gamma_2 + 6 \sum_{n=1}^{\infty} \frac{(H_{n-1} - \gamma - \log n) \log n}{n}. \quad (2.3)$$

Consequently the cubic finite part and the first strict harmonic coefficient are equivalent modulo ordinary zeta and Stieltjes constants. This theorem does not assert that the harmonic coefficient is a finite combination of ordinary zeta constants.

2.1 A global partial-fraction identity

For $n \geq 0$ put

$$b_n = H_n - \gamma, \quad a_n = \zeta(2) + H_n^{(2)} - b_n^2, \quad H_n^{(r)} = \sum_{j=1}^n j^{-r},$$

with $H_0^{(r)} = 0$. Set

$$c_0 = -\gamma^3 + 6\gamma\zeta(2) - 3\zeta(3).$$

Lemma 2.2 (Normally convergent cubic partial fractions). *For $z \notin \{0, -1, -2, \dots\}$,*

$$\begin{aligned} \psi(z)^3 = & -\frac{1}{z^3} - \frac{3\gamma}{z^2} + \frac{3(\zeta(2) - \gamma^2)}{z} + c_0 \\ & + \sum_{n=1}^{\infty} \left\{ -((z+n)^{-3} - n^{-3}) + 3b_n((z+n)^{-2} - n^{-2}) \right. \\ & \left. + 3a_n((z+n)^{-1} - n^{-1}) \right\}. \end{aligned} \quad (2.4)$$

The series converges normally on every compact subset avoiding the poles.

Proof. At $z = -n + \varepsilon$, the recurrence of ψ gives

$$\psi(-n + \varepsilon) = -\varepsilon^{-1} + b_n + (\zeta(2) + H_n^{(2)})\varepsilon + (H_n^{(3)} - \zeta(3))\varepsilon^2 + O(\varepsilon^3).$$

The cubic principal part at $-n$ is therefore

$$P_n(z) = -(z+n)^{-3} + 3b_n(z+n)^{-2} + 3a_n(z+n)^{-1}.$$

The constant coefficient at zero is c_0 . It is important to justify the entire remainder in a Mittag-Leffler argument, rather than infer it from the principal parts alone. Let $f(z) = \psi(z)^3 - P_0(z)$, which is regular at zero, and integrate

$$\frac{zf(w)}{w(w-z)}$$

over the circles $|w| = N + \frac{1}{2}$, for fixed z and large N . On these circles $f(w) = O(\log^3 N)$ uniformly. In the right half-plane this follows from the usual digamma asymptotic [10, Sections 5.7, 5.11]. In the left half-plane use

$$\psi(w) = \psi(1-w) - \pi \cot(\pi w).$$

The cotangent is uniformly bounded: when $|\Im w| \geq 1/4$ this follows from its elementary exponential formula; in the remaining part of the circle, the real coordinate stays a fixed positive distance from the integers. Thus the contour integral is $O(\log^3 N/N)$ and tends to zero. Its residues at z and 0 are $f(z)$ and $-f(0)$. The residue at $-n$ is $-(P_n(z) - P_n(0))$, either by a direct calculation or by applying the same two-point Cauchy kernel to each principal part. The residue theorem proves $f(z) - f(0) = \sum_{n \geq 1} (P_n(z) - P_n(0))$. Finally $b_n = O(\log n)$ and $a_n = O(\log^2 n)$, while on compact sets $(z+n)^{-1} - n^{-1} = O(n^{-2})$. These estimates prove normal convergence and justify the asserted identity. $\square$

2.2 Reduction to one logarithmic harmonic series

Integrating the normally convergent remainder of (2.4) term by term yields

$$Q = \frac{1}{2} + 3\gamma + c_0 + \sum_{n \geq 1} \left\{ \frac{1}{n^3} + \frac{1}{2(n+1)^2} - \frac{1}{2n^2} - \frac{3b_n}{n^2(n+1)} + 3a_n \left(\log \left(1 + \frac{1}{n} \right) - \frac{1}{n} \right) \right\}. \quad (2.5)$$

Here the finite parts of $-x^{-3}$ and $-3\gamma x^{-2}$ are $1/2$ and 3γ , and the unit-coordinate finite part of x^{-1} is zero. The elementary identities

$$\sum_{n \geq 1} \frac{H_n}{n(n+1)} = \zeta(2), \quad \sum_{n \geq 1} \frac{H_n}{n^2} = 2\zeta(3)$$

give

$$\sum_{n \geq 1} \frac{b_n}{n^2(n+1)} = 2\zeta(3) - \zeta(2) - \gamma(\zeta(2) - 1).$$

For completeness, the second Euler sum follows from

$$\sum_{n \geq 1} \frac{H_n}{n^2} = \int_0^1 \frac{\log x \log(1-x)}{x(1-x)} dx = 2 \int_0^1 \frac{\log x \log(1-x)}{x} dx = 2\zeta(3),$$

using the harmonic generating function, symmetry, and an absolutely convergent geometric-series integration. Consequently (2.5) becomes

$$Q = -\gamma^3 + 9\gamma\zeta(2) + 3\zeta(2) - 8\zeta(3) + 3 \sum_{n \geq 1} a_n \left(\log \left(1 + \frac{1}{n} \right) - \frac{1}{n} \right). \quad (2.6)$$

The crucial cancellation is the exact difference

$$a_n - a_{n-1} = -\frac{2(H_{n-1} - \gamma)}{n}.$$

Write $L_N = \sum_{n=1}^N (H_{n-1} - \gamma) \log n/n$ and $S_N = \sum_{n=1}^N H_n/n^2$. Finite summation by parts gives

$$\begin{aligned} \sum_{n=1}^N a_n \log \left(1 + \frac{1}{n} \right) &= a_N \log(N+1) + 2L_N, \\ \sum_{n=1}^N \frac{a_n}{n} &= (\zeta(2) - \gamma^2)H_N + \gamma(H_N^2 + H_N^{(2)}) - \frac{H_N^3}{3} + H_N H_N^{(2)} - 2S_N + \frac{4}{3}H_N^{(3)}. \end{aligned}$$

The second identity follows by summing the finite differences of H_n^2 , H_n^3 , and $H_n H_n^{(2)}$; it has no limiting interchange hidden in it. Since

$$H_N - \gamma = \log(N+1) + O(N^{-1}), \quad H_N^{(2)} = \zeta(2) + O(N^{-1}), \quad S_N \longrightarrow 2\zeta(3),$$

substitution in (2.6), with all polynomial terms retained before the limit, gives

$$Q = 3\zeta(2) + 6 \lim_{N \rightarrow \infty} \left(L_N - \frac{\log^3 N}{3} \right). \quad (2.7)$$

Now split $H_{n-1} - \gamma = \log n + \psi(n) - \log n$ and use the standard defining limit

$$\gamma_2 = \lim_{N \rightarrow \infty} \left(\sum_{n=1}^N \frac{\log^2 n}{n} - \frac{\log^3 N}{3} \right).$$

Since $\psi(n) - \log n = O(n^{-1})$, the remaining series is absolutely convergent. This proves (2.3).

Define the holomorphic Dirichlet deformation

$$D(s) = \sum_{n \geq 1} \frac{\psi(n) - \log n}{n^s}, \quad \Re s > 0.$$

Normal convergence, including every fixed derivative on compact subsets, follows from $\psi(n) - \log n = O(n^{-1})$. In $\Re s > 1$ we have the exact identity

$$D(s) = Z_2(s, 1) - \gamma\zeta(s) + \zeta'(s).$$

Comparing Laurent coefficients at $s = 1$ gives $D'(1) = \eta + \gamma\gamma_1 + \gamma_2$. Formula (2.3) says $Q = 3\zeta(2) + 6\gamma_2 - 6D'(1)$, proving (2.1). The inclusive-to-strict conversion gives (2.2) and completes the proof of Theorem 2.1.

2.3 A normalized complex antiderivative

Corollary 2.3 (Cubic digamma primitive). *On $\Re z > 0$, with the principal logarithm, the function*

$$\begin{aligned} \mathcal{P}_3(z) = & \frac{1}{2z^2} + \frac{3\gamma}{z} + 3(\zeta(2) - \gamma^2) \log z + c_0 z \\ & + \sum_{n=1}^{\infty} \left\{ \frac{1}{2(n+z)^2} - \frac{1}{2n^2} + \frac{z}{n^3} + 3b_n \left(\frac{1}{n} - \frac{1}{n+z} - \frac{z}{n^2} \right) \right. \\ & \left. + 3a_n \left(\log \left(1 + \frac{z}{n} \right) - \frac{z}{n} \right) \right\} \end{aligned} \quad (2.8)$$

satisfies $\mathcal{P}'_3(z) = \psi(z)^3$. Its asymptotic expansion at zero has constant term zero. In particular, for every real $b > 0$,

$$\text{FP} \int_0^b \psi(x)^3 dx = \mathcal{P}_3(b), \quad \mathcal{P}_3(1) = 3\zeta(2) - 6\eta - 6\gamma\gamma_1.$$

The series and its differentiated series converge normally on compact subsets of the right half-plane. Higher derivatives therefore give identities for derivatives of ψ^3 in the same domain.

Proof. Each brace is the primitive, vanishing at $z = 0$, of the corresponding brace in (2.4). On a fixed compact set, the logarithmic difference is $O(n^{-2})$, the quadratic-pole primitive is $O(n^{-3})$, and the cubic-pole primitive is $O(n^{-4})$. Multiplication by $a_n = O(\log^2 n)$ or $b_n = O(\log n)$ preserves normal convergence. Differentiation recovers the previous lemma. All series terms vanish at zero, so the displayed singular and logarithmic terms leave no constant term. This proves the stated normalization. $\square$

Corollary 2.4 (A polygamma–harmonic hierarchy). *For every integer $r \geq 1$ and $z \notin \{0, -1, -2, \dots\}$,*

$$\begin{aligned} & \sum_{r_1+r_2+r_3=r} \frac{r!}{r_1!r_2!r_3!} \psi^{(r_1)}(z) \psi^{(r_2)}(z) \psi^{(r_3)}(z) \\ & = (-1)^r r! \sum_{n=0}^{\infty} \left\{ -\frac{(r+1)(r+2)}{2(n+z)^{r+3}} + \frac{3(r+1)b_n}{(n+z)^{r+2}} + \frac{3a_n}{(n+z)^{r+1}} \right\}. \end{aligned} \quad (2.9)$$

Here $\psi^{(0)} = \psi$. The series converges normally away from the poles. At $z = 1$ the left side is a finite polynomial in $\gamma, \zeta(2), \dots, \zeta(r+1)$, using $\psi^{(k)}(1) = (-1)^{k+1} k! \zeta(k+1)$ for $k \geq 1$.

Proof. Differentiate (2.4) r times. The constant subtractions disappear and the derivative series is bounded on each compact set by a constant times $\sum n^{-r-1}(1 + \log^2 n)$. The product rule gives the left side. The $n = 0$ summand includes the principal part at zero since $b_0 = -\gamma$ and $a_0 = \zeta(2) - \gamma^2$. The special values follow from the classical Hurwitz–polygamma identity [9, Section 25.11]. $\square$

2.4 A short certified evaluation route

For integers $N \geq 2$ and $K \geq 0$ define

$$S_{N,K} = \sum_{n=1}^{N-1} \frac{(H_{n-1} - \gamma - \log n) \log n}{n} + \frac{1}{2} \zeta'(2, N) + \sum_{k=1}^K \frac{B_{2k}}{2k} \zeta'(2k+1, N).$$

The Binet formula for ψ [10, Section 5.9] and the finite geometric expansion of $(n^2 + t^2)^{-1}$ give the rigorous truncation estimate

$$\left| S_{N,K} - \sum_{n \geq 1} \frac{(\psi(n) - \log n) \log n}{n} \right| \leq \frac{|B_{2K+2}|}{2K+2} (-\zeta'(2K+3, N)). \quad (2.10)$$

Indeed the digamma asymptotic remainder is bounded by the magnitude of its first omitted Bernoulli term for positive real arguments, and the logarithmic tail is exactly a Hurwitz-zeta derivative. The supplied Arb script evaluates every displayed quantity with ball arithmetic and adds this analytic tail bound. This certifies the arithmetic identity numerically without replacing its proof.

Status. The harmonic zeta function, its Stieltjes coefficients, and the Mittag–Leffler method are classical. The cubic–harmonic bridge and its coordinate consequence were derived here and were not found in the inspected repository reports. No worldwide priority claim is made.

3 The diagonal Tornheim derivative in harmonic coordinates

The incoming coincident report relates the digamma cube to a third diagonal Tornheim derivative. We give the relevant derivation, with its finite normalization visible, and then insert the new bridge. This separates the earlier spectral identity from the additional harmonic reduction proved here.

Write

$$\omega(t) = \mathcal{T}(t, t, t), \quad \Omega_3 = \omega'''(0), \quad E(t) = \Gamma(1+t)e^{-Lt}, \quad h(t) = \zeta(1-t) + \frac{1}{t}.$$

Here $\mathcal{T}$ is the continuation of $\sum_{m,n \geq 1} m^{-A} n^{-B} (m+n)^{-C}$, and the diagonal is restricted before taking the limit at zero. The third derivative is one-variable data; joint holomorphy at the origin is not assumed.

3.1 The completed cubic generating function

Set

$$g(t, x) = \zeta(1-t, x) + \frac{1}{t} = \sum_{m \geq 0} \gamma_m(x) \frac{t^m}{m!}.$$

Near $x = 0$ its exact Hurwitz splitting yields

$$g(t, x) = x^{t-1} + h(t) - (1-t)\zeta(2-t)x + O(x^2). \quad (3.1)$$

After cubing, the only local monomials whose Mellin denominators vanish at $t = 0$ are

$$3h(t)^2 x^{t-1}, \quad -3(1-t)\zeta(2-t)x^{2t-1}.$$

Their complete spectral integrals are $3h(t)^2/t$ and $-3(1-t)\zeta(2-t)/(2t)$. The complete numerators, rather than just their values at zero, have to be removed to obtain the unit-coordinate finite parts.

For $\Re t > 1$, the absolutely convergent Hurwitz Fourier expansion has coefficients

$$\zeta(\widehat{1-t, \cdot})(n) = \Gamma(t)(2\pi|n|)^{-t} e^{-\pi i t \operatorname{sgn}(n)/2}, \quad n \neq 0.$$

The constant coefficient vanishes. In the product of three series, the condition $n_1 + n_2 + n_3 = 0$ has six sign cones; on each, the largest magnitude is the sum of the other two. Consequently

$$\begin{aligned} \int_0^1 \zeta(1-t, x)^3 dx &= 6\Gamma(t)^3 (2\pi)^{-3t} \cos(\pi t/2) \omega(t), \\ \int_0^1 \zeta(1-t, x)^2 dx &= 2\Gamma(t)^2 (2\pi)^{-2t} \zeta(2t). \end{aligned} \quad (3.2)$$

The classical Hurwitz–Tornheim connection is recorded, with its literature attribution, in the coincident report [2]. Finite endpoint Taylor subtraction continues both sides meromorphically.

Lemma 3.1 (Completed diagonal kernel). *The expression*

$$\begin{aligned} F(t) &= \frac{6E(t)^3 \cos(\pi t/2) \omega(t) + 6E(t)^2 \zeta(2t) + 1}{t^3} \\ &\quad - \frac{3h(t)^2}{t} + \frac{3(1-t)\zeta(2-t)}{2t} \end{aligned} \quad (3.3)$$

is holomorphic near zero and satisfies $F(0) = -Q$.

Proof. Expand g^3 and use (3.2) and the vanishing one-factor integral. This gives the fraction on the first line as the meromorphic integral of g^3 . To justify its completion, split the x integral at any $b \in (0, 1)$. Subtract finitely many Taylor monomials of the smooth factor in (3.1) on $(0, b)$. The remainder and all fixed parameter derivatives are integrable uniformly in a small t disk. A local monomial with exponent $kt - 1$ has spectral integral $b^{kt}/(kt)$, while the generating function of its coordinate finite parts is $(b^{kt} - 1)/(kt)$. Their difference is exactly $1/(kt)$. All nonresonant denominators remain nonzero at zero. The two resonant terms displayed above therefore give precisely the second line of (3.3). The resulting Taylor coefficients are the finite parts of those of g^3 . At zero, $g(0, x) = \gamma_0(x) = -\psi(x)$, hence $F(0) = -Q$. $\square$

Solving (3.3) for ω also proves its holomorphy at zero, since the coefficient $E(t)^3 \cos(\pi t/2)$ is nonzero there. This argument does not infer a joint regularity statement from a ray restriction.

3.2 An explicit reduction of the third diagonal derivative

Use the classical values $\zeta(0) = -1/2$, $\zeta'(0) = -L/2$ and

$$E(t) = \exp \left[-(\gamma + L)t + \frac{\zeta(2)}{2} t^2 - \frac{\zeta(3)}{3} t^3 + O(t^4) \right].$$

The coefficients of t^{-3}, t^{-2}, t^{-1} in (3.3) give

$$\omega(0) = \frac{1}{3}, \quad \omega'(0) = L, \quad \omega''(0) = L^2 - 4\zeta''(0). \quad (3.4)$$

For clarity, after the first two substitutions the coefficient of t^2 in the numerator is

$$3\omega''(0) + 3\gamma^2 - 3L^2 - \frac{3}{2}\zeta(2) + 12\zeta''(0),$$

and the remaining two terms contribute $-3\gamma^2 + 3\zeta(2)/2$ to the residue. The numerator coefficient at t^3 , after (3.4), is

$$\Omega_3 + 8\zeta'''(0) + 12(\gamma + L)\zeta''(0) + L^3 + 6\gamma L^2 - 5\gamma^3 + \frac{3}{2}\gamma\zeta(2).$$

Since $h(t) = \gamma + \gamma_1 t + O(t^2)$, the other constant terms are $-6\gamma\gamma_1 - 3(\zeta(2) + \zeta'(2))/2$. Thus the earlier cubic identity is recovered:

$$\begin{aligned} Q = & -\Omega_3 - 8\zeta'''(0) - 12(\gamma + L)\zeta''(0) - L^3 - 6\gamma L^2 + 5\gamma^3 \\ & + 6\gamma\gamma_1 - \frac{3}{2}\gamma\zeta(2) + \frac{3}{2}(\zeta(2) + \zeta'(2)). \end{aligned} \quad (3.5)$$

Theorem 3.2 (Harmonic coordinate for the cubic diagonal). *The third derivative of the specified diagonal continuation is*

$$\begin{aligned} \Omega_3 = & 6\eta(1) - 8\zeta'''(0) - 12(\gamma + L)\zeta''(0) - L^3 - 6\gamma L^2 + 5\gamma^3 \\ & + 12\gamma\gamma_1 - \frac{3}{2}\gamma\zeta(2) - \frac{3}{2}\zeta(2) + \frac{3}{2}\zeta'(2). \end{aligned} \quad (3.6)$$

In particular, (2.3) gives an absolutely convergent generalized-harmonic evaluation of this derivative.

Proof. Substitute (2.1) into (3.5) and solve for Ω_3 . □

The independently certified arithmetic evaluation in the package gives

$$\begin{aligned} Q &= 1.96626273439153434066437533841163142382586897649\dots, \\ \eta(1) &= 0.53678702435841856750360788434881167385105703307\dots, \\ \Omega_3 &= 86.6578666011000736529032768587621836100797302472\dots \end{aligned}$$

The first harmonic coefficient is therefore sufficient for the diagonal cubic problem. A finite expression for that coefficient in a specified algebra of ordinary zeta, Gamma and Stieltjes constants remains an arithmetic research question.

4 The complete cubic directional layer of Tornheim zeta

Define the uncolored Mordell–Tornheim function in its absolute convergence region by

$$\mathcal{T}(A, B, C) = \sum_{m, n \geq 1} \frac{1}{m^A n^B (m+n)^C}. \quad (4.1)$$

For instance, sufficiently large positive real parts suffice. Its meromorphic continuation is symmetric in A, B , but is not jointly holomorphic at the origin. We form a restriction away from its polar hyperplanes before taking a one-variable limit.

Let

$$W_{a,b,c}(t) = \mathcal{T}(at, bt, ct), \quad \Omega_3 = \left. \frac{d^3}{dt^3} \mathcal{T}(t, t, t) \right|_{t=0}.$$

The two admissibility conditions are

$$(a+c)(b+c) \neq 0. \quad (4.2)$$

They allow complex and unequal slopes, including negative slopes.

4.1 Statement of the formula

Put

$$\begin{aligned}\sigma &= a + b + c, \\ \mathcal{B} &= \frac{c(c^2 + ab - a^2 - b^2)}{(a + c)(b + c)},\end{aligned}\tag{4.3}$$

$$\begin{aligned}\Delta &= a^2 + b^2 - c(a + b), \\ J_3 &= W_{1,0,2}'''(0), \\ \kappa &= J_3 + \frac{35}{2}\zeta'''(0) + 6L\zeta''(0) - 9\zeta'''(-1).\end{aligned}\tag{4.4}$$

The constant J_3 is a fixed Euler double-zeta directional derivative: $\mathcal{T}(t, 0, 2t) = Z_2(2t, t; 1)$. An independent convergent integral for κ is given below.

Theorem 4.1 (All admissible cubic directions). *Under (4.2), $W_{a,b,c}$ is holomorphic at zero and*

$$\begin{aligned}W_{a,b,c}'''(0) &= abc\Omega_3 - \frac{c\Delta}{2}\kappa \\ &\quad + \left(8abc - \frac{(a+c)^3 + (b+c)^3}{2}\right)\zeta'''(0) \\ &\quad - \frac{3}{2}((a+b)(ab+c^2) - 4abc)L\zeta''(0) + \sigma^2\mathcal{B}\zeta'''(-1).\end{aligned}\tag{4.5}$$

Thus every third directional derivative belongs to the span of $\Omega_3, \kappa, \zeta'''(0), L\zeta''(0), \zeta'''(-1)$ over the field of rational functions of the slopes. By the cubic bridge, Ω_3 can be replaced by $\eta(1)$ and explicit ordinary zeta and Stieltjes data.

Every coefficient is homogeneous of degree three. This agrees with $W_{ra,rb,rc}(t) = W_{a,b,c}(rt)$. The formula contains no surviving Euler-constant or Gamma-expansion coefficients in addition to those already packaged into Ω_3 and κ .

4.2 A holomorphic remainder with exactly two visible polar divisors

For $x > 0$ set $F(A, B; x) = \text{Li}_A(e^{-x})\text{Li}_B(e^{-x})$. Jonquière's formula gives, locally uniformly for A near zero,

$$\text{Li}_A(e^{-x}) = \Gamma(1-A)x^{A-1} + \sum_{j \geq 0} \zeta(A-j) \frac{(-x)^j}{j!}, \quad 0 < x < 2\pi.\tag{4.6}$$

Subtract the six local terms

$$\begin{aligned}S(A, B; x) &= \Gamma(1-A)\Gamma(1-B)x^{A+B-2} + \Gamma(1-A)\zeta(B)x^{A-1} \\ &\quad + \Gamma(1-B)\zeta(A)x^{B-1} - \Gamma(1-A)\zeta(B-1)x^A \\ &\quad - \Gamma(1-B)\zeta(A-1)x^B + \zeta(A)\zeta(B).\end{aligned}\tag{4.7}$$

Define

$$\begin{aligned}H(A, B, C) &= \int_0^1 x^{C-1}(F(A, B; x) - S(A, B; x)) \, dx + \int_1^\infty x^{C-1}F(A, B; x) \, dx \\ &\quad + \frac{\Gamma(1-A)\Gamma(1-B)}{A+B+C-2} + \frac{\Gamma(1-A)\zeta(B)}{A+C-1} + \frac{\Gamma(1-B)\zeta(A)}{B+C-1}.\end{aligned}\tag{4.8}$$

All denominators on the last line are nonzero near the origin. On a sufficiently small closed parameter polydisc, the first integrand and every fixed parameter derivative are bounded at zero by $x^{-\delta}(1 + |\log x|^N)$ for a $\delta < 1$; the second integral has exponential decay. Thus H is jointly holomorphic and symmetric in A, B . Termwise integration and meromorphic continuation yield

$$\mathcal{T}(A, B, C) = \frac{1}{\Gamma(1+C)} \left\{ \zeta(A)\zeta(B) - \frac{C\Gamma(1-A)\zeta(B-1)}{A+C} - \frac{C\Gamma(1-B)\zeta(A-1)}{B+C} + CH(A, B, C) \right\}. \quad (4.9)$$

This also proves removability on every admissible ray. The six-term subtraction and the decomposition are retained from the incoming coincident report [2]; the cubic determination below is the additional step.

4.3 Exact slices determine the constant and linear remainder

Two elementary restrictions provide all the lower coefficients:

$$\mathcal{T}(0, 0, t) = \zeta(t-1) - \zeta(t), \quad (4.10)$$

$$\mathcal{T}(t, 0, t) = \frac{\zeta(t)^2 - \zeta(2t)}{2}. \quad (4.11)$$

For the first, count positive pairs with $m+n=k$. For the second, use $\mathcal{T}(A, 0, C) = Z_2(C, A; 1)$ and the exact stuffle product

$$\mathcal{T}(A, 0, C) + \mathcal{T}(C, 0, A) = \zeta(A)\zeta(C) - \zeta(A+C). \quad (4.12)$$

Both identities are first proved in domains of absolute convergence and then extended meromorphically. A fixed $C=0$ substitution alone would not determine the normal derivative of CH ; this is why (4.11) supplies real additional information.

Write

$$\begin{aligned} H(A, B, C) &= h_0 + h_A(A+B) + h_C C \\ &\quad + h_{20}(A^2 + B^2) + h_{11}AB + h_{10c}(A+B)C + h_{02}C^2 + O((|A| + |B| + |C|)^3). \end{aligned} \quad (4.13)$$

Here h_C denotes the coefficient of the linear term in C ; h_{02} is its quadratic coefficient. Direct comparison in (4.10)–(4.11) gives

$$h_0 = \zeta'(-1) + \frac{L}{2} - \frac{5\gamma}{12}, \quad (4.14)$$

$$h_C = \frac{\zeta''(-1) - \zeta''(0)}{2} - \gamma \left(\zeta'(-1) + \frac{L}{2} \right) + \frac{5}{24}(\gamma^2 + \zeta(2)), \quad (4.15)$$

$$h_A = \frac{L^2}{8} - \frac{\zeta''(0)}{2} + \gamma\zeta'(-1) - \frac{\gamma L}{4} - \frac{\gamma^2 + \zeta(2)}{24}. \quad (4.16)$$

For example, the axis gives $\Gamma(1+t)(\zeta(t-1) - \zeta(t)) = 5/12 + tH(0, 0, t)$, which proves the first two lines. On the symmetric slice the coefficient of t^2 then gives the third line.

Substitution now proves the complete lower layer

$$\begin{aligned} W(0) &= \frac{1}{4} + \frac{c}{12(a+c)} + \frac{c}{12(b+c)}, \\ W'(0) &= \frac{a+b+2c}{4}L + \mathcal{B}\zeta'(-1), \\ W''(0) &= \frac{2ab+c(a+b)}{4}L^2 - \frac{a^2+b^2+2c(a+b)+2c^2}{2}\zeta''(0) + \sigma\mathcal{B}\zeta''(-1). \end{aligned} \quad (4.17)$$

These agree with the earlier report. In particular, $\omega(0) = 1/3$, $\omega'(0) = L$, and $\omega''(0) = L^2 - 4\zeta''(0)$ follow here without using the unknown cubic moment.

4.4 Proof of the cubic formula

At quadratic degree, the symmetric slice fixes $h_{20} + h_{10c}$, not the two terms separately. This distinction is the main structural point. The axis fixes h_{02} ; the diagonal then fixes h_{11} , leaving exactly one more coefficient, determined by J_3 .

Here are explicit finite coefficient formulas which also make the algebra completely reproducible. Set $V = h_{20} + h_{10c}$ and define the cubic Taylor polynomials

$$\begin{aligned} O(t) &= \frac{1}{3} + Lt + \frac{L^2 - 4\zeta''(0)}{2}t^2 + \frac{\Omega_3}{6}t^3, \\ D(t) &= \frac{7}{18} + \left(\frac{5}{4}L + \zeta'(-1)\right)t + \left(\frac{L^2}{4} - \frac{13}{4}\zeta''(0) + \frac{3}{2}\zeta''(-1)\right)t^2 + \frac{J_3}{6}t^3. \end{aligned}$$

Comparison on the four slices gives

$$\begin{aligned} h_{02} &= [t^3]\Gamma(1+t)(\zeta(t-1) - \zeta(t)), \\ V + h_{02} &= [t^3]\left\{\frac{\Gamma(1+t)}{2}(\zeta(t)^2 - \zeta(2t)) + \frac{\zeta(t)}{2} - \frac{\Gamma(1-t)}{24} + \zeta(t-1)\right\}, \\ h_{11} + 2V + h_{02} &= [t^3]\left\{\Gamma(1+t)O(t) - \zeta(t)^2 + \Gamma(1-t)\zeta(t-1)\right\}, \\ -2h_{20} + 4V + 8h_{02} &= [t^3]\left\{\Gamma(1+2t)D(t) + \frac{\zeta(t)}{2} - \frac{\Gamma(1-t)}{18} + \zeta(t-1)\right\}. \end{aligned} \quad (4.18)$$

The coefficient system is triangular with nonzero diagonal. Thus it determines all four quadratic coefficients for any assigned Ω_3, J_3 .

Use

$$\Gamma(1 \pm t) = 1 \mp \gamma t + \frac{\gamma^2 + \zeta(2)}{2}t^2 \mp \frac{\gamma^3 + 3\gamma\zeta(2) + 2\zeta(3)}{6}t^3 + O(t^4)$$

and the ordinary Taylor expansions of ζ at 0 and -1 . Substitute (4.18) into (4.9). The coefficients of Ω_3 and J_3 are respectively abc and $-c\Delta/2$. The remaining nonzero coefficients are

$$\begin{aligned} [\zeta'''(0)]: & 8abc - \frac{(a+c)^3 + (b+c)^3}{2} - \frac{35}{4}c\Delta, \\ [L\zeta''(0)]: & -\frac{3}{2}((a+b)(ab+c^2) - 4abc) - 3c\Delta, \\ [\zeta'''(-1)]: & \sigma^2\mathcal{B} + \frac{9}{2}c\Delta. \end{aligned}$$

All coefficients involving $\gamma, \zeta(2), \zeta(3)$ or $\zeta'(-1), \zeta''(-1)$ cancel. Absorbing the three multiples of $c\Delta$ into (4.4) proves Theorem 4.1. This is a finite algebraic proof; the delivered symbolic certificate independently verifies the exact rational identity with all constants treated as formal symbols.

4.5 An independent convergent integral for the second coordinate

Define

$$\begin{aligned}
 Q_2(x) &= \partial_s^2 \text{Li}_s(e^{-x}) \Big|_{s=0} = \sum_{n \geq 1} (\log n)^2 e^{-nx}, \\
 P(x) &= (\log x + \gamma)^2 + \zeta(2), \\
 S_2(x) &= \left(x^{-2} - \frac{1}{2x} + \frac{1}{12} \right) P(x) + \zeta''(0) \left(x^{-1} - \frac{1}{2} \right) - \zeta''(-1), \\
 I_2 &= \int_0^1 \left(\frac{Q_2(x)}{e^x - 1} - S_2(x) \right) \frac{dx}{x} + \int_1^\infty \frac{Q_2(x)}{x(e^x - 1)} dx.
 \end{aligned} \tag{4.19}$$

Both integrals converge ordinarily: the first integrand is $O(1 + \log^2 x)$; the second is $O(e^{-3x}/x)$.

Proposition 4.2 (Mellin coordinate). *The quadratic coefficient h_{20} satisfies*

$$h_{20} = \frac{\gamma}{4} + \frac{3}{8} - \frac{\zeta''(0)}{2} + \frac{I_2}{2}, \tag{4.20}$$

and

$$\kappa = -12h_{20} - \frac{\gamma^3}{6} - \frac{\gamma\zeta(2)}{2} + 3\gamma\zeta''(0) + 6\gamma\zeta''(-1) - \frac{\zeta(3)}{3} - 3\zeta'''(0) - 2\zeta'''(-1). \tag{4.21}$$

Proof. Differentiate (4.8) twice in A at $(0, 0, 0)$. The product derivative is $Q_2(x)/(e^x - 1)$, and the derivative of the six subtractions is $S_2(x)$. Differentiation under the subtracted integrals follows from the local bounds used above. The rational last line of (4.8), on $B = C = 0$, is

$$\Gamma(1 - A) \left(\frac{1}{A - 2} - \frac{1}{2(A - 1)} \right) - \zeta(A).$$

Its coefficient of A^2 is $\gamma/4 + 3/8 - \zeta''(0)/2$, proving (4.20). Expanding the last line of (4.18) and using (4.4) gives (4.21). $\square$

The independent integral evaluation gives

$$\begin{aligned}
 I_2 &= -0.2860481161626797704387366252971602\dots, \\
 \kappa &= -3.0226121641572464811701683702659737\dots
 \end{aligned}$$

These displayed digits are numerical diagnostics, not an arithmetic characterization. The definition (4.19) supplies such a characterization without referring back to an unknown generic ray.

4.6 Coordinate cancellation and exact examples

In the formal coordinate system of Theorem 4.1, the κ term disappears exactly when

$$c(a^2 + b^2 - c(a + b)) = 0. \tag{4.22}$$

For positive a, b , the choice $c = (a^2 + b^2)/(a + b)$ gives a continuum of asymmetric directions with only the harmonic coordinate $\eta(1)$ remaining. For fixed real c , the nontrivial locus can equivalently be written $(a - c/2)^2 + (b - c/2)^2 = c^2/2$.

The Ω_3 term disappears when $abc = 0$. With $c \neq 0$, both displayed nonclassical coefficients vanish only on the axis $(0, 0, c)$ and the rays $(c, 0, c)$ or $(0, c, c)$. With $c = 0$, admissibility requires

$ab \neq 0$ and the restriction equals $\zeta(at)\zeta(bt)$. These are coefficient-elimination statements, not assertions that other numerical cancellations are impossible.

Some explicit specializations are

$$W_{1,2,3}'''(0) = 6\Omega_3 + 6\kappa - \frac{93}{2}\zeta'''(0) - \frac{27}{2}L\zeta''(0) + \frac{162}{5}\zeta'''(-1), \quad (4.23)$$

$$W_{2,3,1}'''(0) = 6\Omega_3 - 4\kappa + \frac{5}{2}\zeta'''(0) - \frac{33}{2}L\zeta''(0) - 18\zeta'''(-1), \quad (4.24)$$

$$W_{1,1,2}'''(0) = 2\Omega_3 + 2\kappa - 11\zeta'''(0) - 3L\zeta''(0) + \frac{32}{3}\zeta'''(-1),$$

$$W_{0,0,1}'''(0) = \zeta'''(-1) - \zeta'''(0),$$

$$W_{1,0,1}'''(0) = -\frac{9}{2}\zeta'''(0) - \frac{3}{2}L\zeta''(0). \quad (4.25)$$

The last two also follow directly from the exact slices.

Remark 4.3 (Why two coordinates are sufficient, and what this does not prove). Modulo the already determined constant and linear coefficients of H , the axis and symmetric slice leave two formal quadratic degrees of freedom. The diagonal and $(1, 0, 2)$ slices fix them separately. Their contributions to generic cubic rays are independent polynomials abc and $c\Delta$. This proves uniqueness in the specified formal reconstruction problem. It does not prove that $\eta(1)$ and κ are independent over any arithmetic constant field.

5 Transverse curved directions and a sharp fourth-jet correction

The cubic coordinate formula survives a nonlinear change of spectral path. The path contributes an explicitly classical correction, while the coefficients of Ω_3 and κ depend only on its tangent.

Using (4.14)–(4.16), define

$$K(A, B, C) = \frac{1}{\Gamma(1+C)} \left\{ \zeta(A)\zeta(B) - \frac{C\Gamma(1-A)\zeta(B-1)}{A+C} - \frac{C\Gamma(1-B)\zeta(A-1)}{B+C} + C(h_0 + h_A(A+B) + h_C C) \right\}. \quad (5.1)$$

Although K is meromorphic in three variables, its restriction to each of the paths in the following theorem is holomorphic.

Theorem 5.1 (Cubic transport along a spectral curve). *Let $A(t), B(t), C(t)$ be analytic near zero, vanish at zero, and have first derivatives a, b, c satisfying (4.2). Write $\mathcal{W}(t) = \mathcal{T}(A(t), B(t), C(t))$. Then*

$$\mathcal{W}'''(0) = W_{a,b,c}'''(0) + 6[t^3](K(A(t), B(t), C(t)) - K(at, bt, ct)). \quad (5.2)$$

The correction uses only ordinary Gamma and zeta derivatives and the Taylor coefficients of A, B, C through degree four. In particular, the coefficients of Ω_3 and κ remain abc and $-c\Delta/2$.

Proof. Write the quadratic homogeneous part of (4.13) as H_2 . Formula (4.9) gives

$$\mathcal{T} - K = \frac{C}{\Gamma(1+C)} H_2(A, B, C) + O((|A| + |B| + |C|)^4).$$

After substitution of the curve, its coefficient of t^3 is $cH_2(a, b, c)$, exactly the coefficient on the tangent ray. Subtracting the two restrictions proves (5.2).

The only possible need for one additional path coefficient comes from $C/(A + C)$ and $C/(B + C)$. Remove their common factor t . The new denominators have nonzero constant terms, so the coefficient of t^3 in each ratio uses the original curves only through degree four. All the remaining compositions use at most degree three. This proves the finite-jet assertion and holomorphy of the restrictions. $\square$

5.1 A correction involving only the lower directional layer

The correction can be made smaller than the Gamma expression (5.1) suggests. Put $v = (a, b, c)$ and write the path as

$$(A(t), B(t), C(t)) = tv + t^2\alpha + t^3\beta + t^4\delta + O(t^5).$$

Let $f_j(v) = W_v^{(j)}(0)$ for $j = 0, 1, 2$, explicitly given by (4.17). Denote ordinary differentials in the slope variables by D . Then

$$\begin{aligned} \mathcal{W}'''(0) - W_v'''(0) &= 6Df_0(v)[\delta] + 6D^2f_0(v)[\alpha, \beta] + D^3f_0(v)[\alpha, \alpha, \alpha] \\ &\quad + 6Df_1(v)[\beta] + 3D^2f_1(v)[\alpha, \alpha] + 3Df_2(v)[\alpha]. \end{aligned} \quad (5.3)$$

In particular the correction belongs to the span of $1, L, L^2, \zeta'(-1), \zeta''(0), \zeta''(-1)$ with rational functions of the path coefficients. No third zeta derivative or additional Gamma constant is needed.

To prove (5.3), cancel the common factor t in the two polar ratios in (4.9). The function $(t, v) \mapsto \mathcal{T}(ta, tb, tc)$ is then jointly holomorphic near $(0, v)$ for admissible v . Its Taylor series is $\sum_{j \geq 0} f_j(v)t^j/j!$. Substitute $v + \alpha t + \beta t^2 + \delta t^3 + O(t^4)$ for v and take the coefficient of t^3 . The ordinary multivariate Taylor formula gives exactly (5.3).

Proposition 5.2 (The fourth jet is genuinely needed). *Fix admissible a, b, c with $c \neq 0$. Replacing the straight path (at, bt, ct) by $(at + \varepsilon t^4, bt, ct)$ changes its cubic derivative by exactly*

$$-\frac{c\varepsilon}{2(a+c)^2}. \quad (5.4)$$

Here ε is the coefficient of t^4 , not the fourth derivative.

Proof. Only the ratio $C/(A + C)$ in (5.1) changes at degree three. Its multiplier at the origin is $-\zeta(-1) = 1/12$. The ratio changes by $-c\varepsilon t^3/(a+c)^2 + O(t^4)$. Multiplication by $3! = 6$ gives (5.4). $\square$

For example, the curve $(t + \varepsilon t^4, t, t)$ has cubic derivative $\Omega_3 - \varepsilon/8$, even though its first three path derivatives agree with those of the diagonal.

This theorem concerns transverse curves. If a path lies in $A + C = 0$ or $B + C = 0$, or is tangent to one of those divisors, a separate restriction or normal finite-part prescription is needed. The theorem does not assign a value by substituting a zero denominator into (4.5).

6 Arbitrary depth with mixed integer inner orders

Question 11 of the preceding directional report asks for a finite nested-polylogarithm reduction when the inner integer orders have both signs. Theorem 6.1 supplies such a reduction while keeping the outer order complex. If exactly q inner orders are positive, its output has depth at most $q + 1$, regardless of how many nonpositive orders occur. Its coefficients are finite rational

operations, root-of-unity filters, and generalized-harmonic polynomials. Repeated poles are included directly, and coincident colors are evaluated by an explicit polynomial summation rule.

The shuffle algebra of iterated integrals and the reduction of rational functions times hyperlogarithms are established tools; see Panzer [11, Sections 2.1–2.3]. Colored depth-two Tornheim reductions are also classical [12]. The contribution here is the explicit convolution formula with unrestricted outer order, its depth bound, its coincident-color reduction rule, and the accompanying meromorphic and normal-derivative prescription. No claim of literature priority for the underlying shuffle or rational-function algorithms is made.

6.1 Definitions and finite input data

Let $s_j \in \mathbb{Z}$, $p_j \in \mathbb{Z}_{>0}$, and $0 < |X_j| \leq 1$. Define

$$Z_{\mathbf{s}, \mathbf{p}}(C; \mathbf{X}) = \sum_{m_1, \dots, m_d \geq 1} \frac{\prod_{j=1}^d X_j^{m_j} m_j^{-s_j}}{(p_1 m_1 + \dots + p_d m_d)^C}. \quad (6.1)$$

The half-plane $\Re C > d + \sum_j \max(0, -s_j)$ is sufficient for absolute convergence. Every power of a positive integer uses its real logarithm. Put

$$I_+ = \{j : s_j > 0\}, \quad I_- = \{j : s_j \leq 0\}, \quad q = |I_+|,$$

and assume for the main theorem that both sets are nonempty. Factor the ordinary coefficient generating function as

$$\sum_{n \geq 1} r_n t^n = R(t)P(t), \quad R(t) = \prod_{j \in I_-} \text{Li}_{s_j}(X_j t^{p_j}), \quad P(t) = \prod_{j \in I_+} \text{Li}_{s_j}(X_j t^{p_j}). \quad (6.2)$$

Then $Z(C) = \sum_{n \geq 1} r_n n^{-C}$ in the stated half-plane.

For each $j \in I_-$ write $a_j = -s_j$. The identity

$$\text{Li}_{-a}(z) = \left(z \frac{d}{dz}\right)^a \frac{z}{1-z}$$

shows that R is rational and bounded at infinity. Its finite partial fractions have the form

$$R(t) = c_\infty + \sum_{\alpha} \sum_{h=1}^{M_\alpha} \frac{c_{\alpha,h}}{(1-\alpha t)^h}, \quad M_\alpha = \sum_{\substack{j \in I_- \\ \alpha^{p_j} = X_j}} (1 - s_j). \quad (6.3)$$

The index α runs over the distinct roots of the indicated equations. Pole cancellation may make some coefficients zero. Even at repeated roots the coefficient formula is simply

$$c_{\alpha,h} = [u^{M_\alpha-h}] u^{M_\alpha} R((1-u)/\alpha). \quad (6.4)$$

The function inside the coefficient extraction is holomorphic at zero. Because I_- is nonempty, $R(0) = 0$, so

$$c_\infty + \sum_{\alpha,h} c_{\alpha,h} = 0. \quad (6.5)$$

Finally set

$$e_{h,r} = [v^r] \prod_{\nu=1}^{h-1} (1 + v/\nu) \quad (0 \leq r < h). \quad (6.6)$$

Thus $e_{h,0} = 1$, $e_{h,1} = H_{h-1}$, and $e_{h,2} = \{H_{h-1}^2 - H_{h-1}^{(2)}\}/2$. In every degree the coefficients follow from

$$\prod_{\nu=1}^{h-1} (1 + v/\nu) = \exp \left(\sum_{k \geq 1} \frac{(-1)^{k+1}}{k} H_{h-1}^{(k)} v^k \right).$$

We use the descending-index convention

$$\text{Li}_{u_1, \dots, u_k}(z_1, \dots, z_k) = \sum_{n_1 > \dots > n_k \geq 1} \frac{z_1^{n_1} \dots z_k^{n_k}}{n_1^{u_1} \dots n_k^{u_k}}. \quad (6.7)$$

The first order will remain variable. Intermediate trailing orders may be nonpositive integers; Lemma 6.2 removes all such orders. For root choices $\mathbf{a} = (a_j)_{j \in I_+}$ with $a_j^{p_j} = X_j$, form the words

$$0^{s_j-1} a_j \quad (j \in I_+).$$

Their shuffle, with each original position retained as a labeled position before equal words are collected, is a finite formal sum $\sum_w \mu_w w$. Every resulting word has exactly q nonzero letters and ends in a nonzero letter; hence it has a unique form

$$w = 0^{k_1-1} b_1 0^{k_2-1} b_2 \dots 0^{k_q-1} b_q, \quad k_i \geq 1. \quad (6.8)$$

The sequence $(b_1, \dots, b_q)$ lists the selected root colors in the order in which they occur in that shuffle. Repeated colors do not alter this rule or suppress their shuffle multiplicities.

6.2 The mixed-sign reduction theorem

Theorem 6.1 (Finite mixed-sign reduction and every normal jet). *Let the preceding data be fixed, with $q \geq 1$ and $I_- \neq \emptyset$. Put $K = \prod_{j \in I_+} p_j^{s_j-1}$. Then*

$$\begin{aligned} Z(C) = K \sum_{\substack{\mathbf{a} \\ a_j^{p_j} = X_j}} \sum_w \mu_w \sum_{\alpha} \sum_{h=1}^{M_{\alpha}} c_{\alpha,h} \sum_{r=0}^{h-1} e_{h,r} \sum_{\ell=0}^r (-1)^{\ell} \binom{r}{\ell} \\ \cdot \text{Li}_{C-r+\ell, k_1-\ell, k_2, \dots, k_q} \left(\alpha, \frac{b_1}{\alpha}, \frac{b_2}{b_1}, \dots, \frac{b_q}{b_{q-1}} \right). \end{aligned} \quad (6.9)$$

This is an identity of meromorphic functions of C on the whole complex plane. It reduces, by finite algebra, to a sum of terms

$$A \text{Li}_{C+v, t_2, \dots, t_k}(z_1, \dots, z_k), \quad v \in \mathbb{Z}, \quad t_i \in \mathbb{Z}_{>0}, \quad k \leq q+1. \quad (6.10)$$

All coefficients and colors are independent of C . Consequently every derivative $\partial_C^N Z$ is obtained by differentiating only the first polylogarithm order in either finite formula. If no X_j equals one, the continuation is entire and these derivatives are ordinary entire functions. When poles occur, the identities and derivatives are meromorphic identities with the pole prescription in Theorem 6.4 below.

Proof. The root filter, valid first as an identity of power series, is

$$\text{Li}_s(Xt^p) = p^{s-1} \sum_{a^p=X} \text{Li}_s(at). \quad (6.11)$$

Indeed the sum of the n th powers of those roots vanishes unless $p \mid n$, and equals $pX^{n/p}$ otherwise. Thus no choice of a particular p th-root branch enters the result.

For completeness, take the iterated-integral forms $\omega_0 = dt/t$ and $\omega_a = a dt/(1 - at)$, and let $\mathcal{H}_w$ denote the corresponding integral from zero with outermost letter first. All words used here end in a nonzero letter, so these integrals require no tangential-base-point regularization. Expansion of the kernels into geometric series proves

$$\mathcal{H}_w(t) = \text{Li}_{k_1, \dots, k_q}(b_1 t, b_2/b_1, \dots, b_q/b_{q-1}). \quad (6.12)$$

Partitioning the product of integration simplices by all total orderings of their integration variables proves the shuffle product, including its multiplicities when letters coincide. Applying this to (6.11) gives

$$P(t) = K \sum_{\mathbf{a}} \sum_w \mu_w \mathcal{H}_w(t).$$

It remains to compute one rational-factor convolution. Write $H(t) = \mathcal{H}_w(t) = \sum_{m \geq 1} h_m t^m$. For $n \geq m$,

$$\begin{aligned} [t^{n-m}](1 - \alpha t)^{-h} &= \alpha^{n-m} \binom{n-m+h-1}{h-1} \\ &= \alpha^{n-m} \sum_{r=0}^{h-1} e_{h,r} \sum_{\ell=0}^r (-1)^\ell \binom{r}{\ell} n^{r-\ell} m^\ell. \end{aligned} \quad (6.13)$$

The diagonal $n = m$ contributes h_n exactly once. The strict part $n > m$, followed by the nested indices already in h_m , is precisely the multiple polylogarithm in (6.9). The diagonal contributions from every rational partial fraction and from $c_\infty H$ have combined coefficient $c_\infty + \sum_{\alpha, h} c_{\alpha, h} = R(0) = 0$. Thus they cancel exactly, leaving the displayed formula.

This reasoning proves an identity of coefficient sequences. All series converge locally normally in C if $|X_j| < 1$: the cumulative colors in every term of (6.9) are $(\alpha, b_1, \dots, b_q)$, all strictly inside the disk, so the coefficients have exponential decay times a polynomial. For boundary colors the same coefficient identity can first be summed in a sufficiently far right half-plane. Lemma 6.2 produces (6.10); each elimination decreases depth by one. The initial depth is $q + 1$, proving the asserted depth bound. The Mellin argument in Theorem 6.4 proves the continuation and the entire assertion. Equality on the original half-plane then implies equality meromorphically everywhere and permits every fixed C derivative. $\square$

Lemma 6.2 (Eliminating a nonpositive trailing order). *Any multiple polylogarithm whose first order is $C + v$ with $v \in \mathbb{Z}$ and whose remaining orders are integers reduces finitely to terms of the form (6.10), with depth no greater than its initial depth. Coincident cumulative colors are handled directly, without division by their difference.*

Proof. Choose any nonpositive trailing order $-M$ at position $j \geq 2$. For fixed neighboring indices put $u = n_{j-1}$ and $l = n_{j+1}$, with $n_{k+1} = 0$ when $j = k$ is last. The sum to eliminate is

$$\sum_{l < n < u} \beta^n n^M, \quad \beta = z_j.$$

For $\beta \neq 1$ define the rational functions and polynomial

$$S_r(\beta) = \left(\beta \frac{d}{d\beta} \right)^r \frac{1}{1 - \beta}, \quad Q_M(x; \beta) = \sum_{r=0}^M \binom{M}{r} x^{M-r} S_r(\beta). \quad (6.14)$$

The geometric-series identity, or its polynomial continuation in β , gives

$$\sum_{l < n < u} \beta^n n^M = \beta^{l+1} Q_M(l+1; \beta) - \beta^u Q_M(u; \beta). \quad (6.15)$$

For example, it follows by subtracting the two tails $\sum_{n \geq v} \beta^n n^M = \beta^v Q_M(v; \beta)$ when $|\beta| < 1$, and then using equality of rational functions. At $\beta = 1$ the formula used by the algorithm is instead

$$\sum_{l < n < u} n^M = \frac{B_{M+1}(u) - B_{M+1}(l+1)}{M+1}, \quad (6.16)$$

where $B_1(x) = x - 1/2$. Both identities hold even when $u = l + 1$; their two boundary terms then cancel.

Expand the boundary polynomials in u or l . A power u^a lowers the preceding order by a , and the factor β^u multiplies the preceding color by β . Similarly, a power l^a lowers the next order and β^l multiplies its color by β . At the last position the lower boundary is a scalar. After removing position j , every term is a polylogarithm of depth one less, with first order still of the form $C +$ an integer. Some adjacent trailing orders may now be nonpositive; repeat the same step. The depth strictly decreases at each step, so the recursion terminates after at most $k - 1$ eliminations along any branch. Every trailing order in the final terms is positive. Equation (6.16) is the exact coincident-color branch; it never requires taking numerically unstable limits of (6.15). $\square$

Remark 6.3 (What the reduction preserves). Only the first polylogarithm order varies with C . The theorem does not authorize differentiating the finite expression with respect to the discrete inner indices s_j . Boundary elimination replaces a pair of adjacent colors by their product; hence its cumulative colors are obtained by deleting entries from $(\alpha, b_1, \dots, b_q)$. In particular, if every original X_j is nontrivial, it cannot introduce a cumulative color equal to one. This observation also explains why equal neighboring colors are harmless while a genuinely uncolored original factor can change the outer spectral poles.

The two unmixed sectors fit the same proof. If $I_- = \emptyset$, the shuffle gives a sum of depth- d polylogarithms with first order $C + k_1$. If $I_+ = \emptyset$, the coefficient identity (6.13) gives the preceding depth-one formula

$$Z(C) = \sum_{\alpha, h} c_{\alpha, h} \sum_{r=0}^{h-1} e_{h, r} \text{Li}_{C-r}(\alpha).$$

Thus every pattern of integer inner orders has a finite reduction.

6.3 Continuation, poles, and normal differentiation

Positive uncolored factors introduce logarithms. Therefore the finite simple-pole list from the all-nonpositive rational sector cannot simply be copied to the mixed sector. The following local rule gives all the poles and their principal coefficients.

Theorem 6.4 (Complete principal-part prescription). *For the integer-order sum (6.1), including the unmixed sectors, put*

$$F(x) = \prod_j \text{Li}_{s_j}(X_j e^{-p_j x}), \quad M = \sum_{\substack{s_j \leq 0 \\ X_j = 1}} (1 - s_j), \quad \ell_+ = \#\{j : s_j > 0, X_j = 1\}.$$

Near $x = 0^+$ there is a convergent logarithmic Laurent expansion

$$F(x) = \sum_{v \geq -M} \sum_{b=0}^{\ell_+} a_{v, b} x^v (\log x)^b. \quad (6.17)$$

The meromorphic continuation of Z has possible poles only at integers $m \leq M$. Its complete principal part at $C = m$, with $\varepsilon = C - m$, is

$$\text{PP}_{C=m} Z(C) = \text{PP}_{\varepsilon=0} \left\{ \frac{1}{\Gamma(m + \varepsilon)} \sum_{b=0}^{\ell_+} \frac{(-1)^b b! a_{-m,b}}{\varepsilon^{b+1}} \right\}. \quad (6.18)$$

Consequently the pole order is at most $\ell_+ + 1$ for positive m , and at most ℓ_+ for nonpositive m . Missing coefficients and canceled principal parts mean removable singularities. If no $X_j = 1$, the function is entire. Every normal derivative is obtained by differentiating the meromorphic function and its displayed principal parts.

Proof. For a nontrivial color the factor is analytic at zero, with expansion

$$\text{Li}_s(X e^{-px}) = \sum_{k \geq 0} \text{Li}_{s-k}(X) \frac{(-px)^k}{k!}. \quad (6.19)$$

For $a \geq 0$ the uncolored nonpositive factor has expansion

$$\text{Li}_{-a}(e^{-px}) = \frac{a!}{(px)^{a+1}} + \sum_{k \geq 0} \zeta(-a - k) \frac{(-px)^k}{k!}. \quad (6.20)$$

For $s \geq 1$ the uncolored positive factor has the integer-order limit of Jonquière's expansion [8, Eq. 25.12.12]:

$$\text{Li}_s(e^{-px}) = \frac{(-px)^{s-1}}{(s-1)!} \{H_{s-1} - \log p - \log x\} + \sum_{\substack{k \geq 0 \\ k \neq s-1}} \zeta(s-k) \frac{(-px)^k}{k!}. \quad (6.21)$$

These convergent local expansions prove (6.17) and provide every needed coefficient by finite multiplication.

In the initial right half-plane, termwise integration gives

$$Z(C) = \frac{1}{\Gamma(C)} \int_0^\infty x^{C-1} F(x) dx. \quad (6.22)$$

The integral over any $[\delta, \infty)$ is entire before multiplication by $1/\Gamma(C)$, since F decreases exponentially at infinity. Subtract successively more terms of (6.17) near zero. The remainder then defines a holomorphic function on successively larger left half-planes, and the subtracted monomials contribute

$$\int_0^1 x^{C+v-1} (\log x)^b dx = \frac{(-1)^b b!}{(C+v)^{b+1}}.$$

Only $v = -m$ can contribute to the principal part at $C = m$. The reciprocal Gamma function is nonzero at positive integers and has a simple zero at each nonpositive integer. This proves (6.18) and the pole-order assertions. If there are no uncolored factors, then $M = \ell_+ = 0$; all potential Mellin poles cancel against those Gamma zeros. Local normal convergence of the subtracted integrals on compact sets also justifies every fixed normal derivative.

To verify continuation of each nested term separately, its coefficient kernel, with the first denominator omitted, is

$$\sum_{n_1 > \dots > n_k \geq 1} \frac{(z_1 t)^{n_1} z_2^{n_2} \dots z_k^{n_k}}{n_2^{t_2} \dots n_k^{t_k}} = \frac{z_1 t}{1 - z_1 t} \text{Li}_{t_2, \dots, t_k}(z_1 z_2 t, z_3, \dots, z_k).$$

For depth one the kernel is just $z_1 t / (1 - z_1 t)$. The positive trailing indices make the right side a rational function times an iterated integral. Induction in its word length gives a convergent finite-logarithmic local expansion at $t = 1$. Applying the same Mellin subtractions with parameter $C + v$ proves its meromorphic continuation. If none of its cumulative colors is one, its kernel is analytic at $t = 1$, and the continuation is entire. As noted after Lemma 6.2, all these cumulative colors come from the original root sets. $\square$

Formula (6.18) is fully finite at every specified pole. For clarity, the Gamma factors needed there can be generated by

$$\begin{aligned} \frac{1}{\Gamma(m + \varepsilon)} &= \frac{1}{(m-1)!} \exp \left\{ (\gamma - H_{m-1})\varepsilon + \sum_{k \geq 2} \frac{(-1)^{k+1}}{k} (\zeta(k) - H_{m-1}^{(k)})\varepsilon^k \right\}, \quad m \geq 1, \\ \frac{1}{\Gamma(-N + \varepsilon)} &= (-1)^N N! \varepsilon \exp \left\{ (\gamma - H_N)\varepsilon + \sum_{k \geq 2} \frac{(-1)^{k+1}}{k} \zeta(k) - H_N^{(k)} \varepsilon^k \right\}. \end{aligned} \quad (6.23)$$

Only finitely many coefficients are required. This controls all principal coefficients through ordinary zeta values, generalized harmonic numbers, integer-order polylogarithms, and logarithms of the weights. It does not assert that the holomorphic Laurent coefficients reduce to depth one.

Inside the unit polydisk the functions are already entire in C . When all boundary colors are nontrivial, their continuation agrees locally uniformly in C with the total-degree Abel limit $X_j \mapsto \rho^{p_j} X_j$, $\rho \uparrow 1$. This follows from the same Taylor subtractions, uniformly near those fixed colors. If uncolored factors occur, an Abel limit outside the convergence half-plane need not exist. In that case the prescribed value is the meromorphic continuation of the fixed-color sum, and a pole requires a Laurent coefficient or other explicitly stated restriction.

6.4 Explicit mixed and weighted identities

For compactness write

$$Z_{s_1, \dots, s_d}(C; z) = \sum_{m_1, \dots, m_d \geq 1} \frac{z^{m_1 + \dots + m_d}}{m_1^{s_1} \dots m_d^{s_d} (m_1 + \dots + m_d)^C}.$$

The following identities hold first for $|z| < 1$, and thereafter with the fixed-color continuation just specified. From $\text{Li}_1(z)^2 = 2 \text{Li}_{1,1}(zt, 1)$ and $\text{Li}_0(z) = -1 + (1 - zt)^{-1}$, one obtains

$$Z_{1,1,0}(C; z) = 2 \text{Li}_{C,1,1}(z, 1, 1). \quad (6.24)$$

The repeated pole

$$\text{Li}_{-1}(zt) = (1 - zt)^{-2} - (1 - zt)^{-1}$$

gives, after the coincident-color elimination,

$$\begin{aligned} Z_{1,1,-1}(C; z) &= 2 \text{Li}_{C-1,1,1}(z, 1, 1) - 2 \text{Li}_{C-1,1}(z, 1) + 2 \text{Li}_{C,1}(z, 1) \\ &\quad + 2 \text{Li}_{C-1}(z) - 2 \text{Li}_C(z). \end{aligned} \quad (6.25)$$

Indeed the intermediate term is $2\{\text{Li}_{C-1,1,1} - \text{Li}_{C,0,1}\}$, and direct summation of the middle index proves

$$\text{Li}_{a,0,1}(z, 1, 1) = \text{Li}_{a-1,1}(z, 1) - \text{Li}_{a,1}(z, 1) - \text{Li}_{a-1}(z) + \text{Li}_a(z).$$

Every normal derivative of (6.25), including at $C = 0$ when $z \neq 1$, follows by replacing each polylogarithm with its corresponding first-order derivative. This gives the whole normal jet, rather than an isolated negative-integer value identity.

A four-variable example has depth at most three, as predicted:

$$\begin{aligned} Z_{1,1,0,0}(C; z) &= 2 \operatorname{Li}_{C-1,1,1}(z, 1, 1) - 2 \operatorname{Li}_{C,1,1}(z, 1, 1) \\ &\quad - 2 \operatorname{Li}_{C-1,1}(z, 1) + 2 \operatorname{Li}_{C,1}(z, 1) + 2 \operatorname{Li}_{C-1}(z) - 2 \operatorname{Li}_C(z). \end{aligned} \quad (6.26)$$

Here the rational factor is $(zt)^2/(1-zt)^2 = 1 - 2/(1-zt) + (1-zt)^{-2}$.

An unequal-weight example, with both root filtering and a color coincidence, is

$$W(C) = \sum_{m,n,k \geq 1} \frac{4^{-m} 3^{-n} 9^{-k}}{m^2 n (2m + n + 2k)^C}. \quad (6.27)$$

It is entire in C . Since

$$\operatorname{Li}_2(t^2/4) = 2\{\operatorname{Li}_2(t/2) + \operatorname{Li}_2(-t/2)\}, \quad \frac{t^2/9}{1-t^2/9} = -1 + \frac{1}{2(1-t/3)} + \frac{1}{2(1+t/3)},$$

the three shuffles of the words $0a$ and $1/3$ give

$$\begin{aligned} W(C) &= \sum_{\substack{a \in \{1/2, -1/2\} \\ \delta \in \{1/3, -1/3\}}} \left\{ \operatorname{Li}_{C,1,2}(\delta, 1/(3\delta), 3a) + \operatorname{Li}_{C,2,1}(\delta, 1/(3\delta), 3a) \right. \\ &\quad \left. + \operatorname{Li}_{C,2,1}(\delta, a/\delta, 1/(3a)) \right\}. \end{aligned} \quad (6.28)$$

All cumulative colors in these twelve terms have modulus less than one, so no continuation convention is needed to differentiate this identity at any C .

The uncolored case of (6.24) illustrates why the pole theorem is necessary. Near zero its Mellin kernel is

$$\operatorname{Li}_0(e^{-x}) \operatorname{Li}_1(e^{-x})^2 = x^{-1} \log^2 x - \frac{1}{2} \log^2 x - \log x + x \left\{ \frac{1}{12} \log^2 x + \frac{7}{12} \log x + \frac{1}{4} \right\} + O(x^2(1+|\log x|^2)).$$

Consequently

$$\begin{aligned} \operatorname{PP}_{C=1} Z_{1,1,0}(C; 1) &= \frac{2}{(C-1)^3} + \frac{2\gamma}{(C-1)^2} + \frac{\gamma^2 - \zeta(2)}{C-1}, \\ \operatorname{PP}_{C=0} Z_{1,1,0}(C; 1) &= -\frac{1}{C^2} + \frac{1-\gamma}{C}, \\ \operatorname{PP}_{C=-1} Z_{1,1,0}(C; 1) &= -\frac{1}{6(C+1)^2} + \frac{3/4 - \gamma/6}{C+1}. \end{aligned} \quad (6.29)$$

In fact there are infinitely many nonpositive poles: the coefficient of $\log^2 x$ in this kernel is exactly $\operatorname{Li}_0(e^{-x})$, whose positive odd powers have nonzero Bernoulli coefficients. This observation concerns the mixed extension; it is not a correction of the earlier theorem restricted to rational generating functions.

6.5 A compatible primitive hierarchy and exact checks

With r_n from (6.2), introduce an additional variable z by $Z(C; z) = \sum_{n \geq 1} r_n z^n n^{-C}$. Equivalently this replaces X_j by $X_j z^{p_j}$. In every term of the finite reduction only the first polylogarithm color acquires the factor z . Thus

$$\int_0^z \partial_C^N Z(C; t) \frac{dt}{t} = \partial_C^N Z(C+1; z), \quad z \frac{d}{dz} \partial_C^N Z(C; z) = \partial_C^N Z(C-1; z). \quad (6.30)$$

The primitive is fixed by its zero value at $z = 0$. These identities follow by local normal convergence in the disk and continue along any common path avoiding the coefficient-function singularities.

The accompanying program performs 624 exact rational checks with SymPy; its source is `verify_mixed_reduction.py`. There are 484 coefficient checks: for eleven independent fixtures, each of the first 22 coefficients is computed directly from the original product and compared both with (6.9) and with the fully eliminated positive-trailing-order expression. The fixtures include four original variables, three positive factors, unequal weights, repeated rational poles, coincident and opposite unit colors, and nonpositive intermediate trailing indices. The remaining 140 checks verify the finite boundary summations for degrees zero through six, including empty intervals and $\beta = 1$. No floating-point tolerance is used. The coefficient checks establish finite algebraic diagnostics; meromorphic continuation and differentiation are supplied by the proofs, not inferred from these tests. The exact outputs and fixture descriptions are recorded in `mixed_reduction_checks.json`.

The remaining question is arithmetic simplification of the resulting positive-depth normal jets. The depth bound is a guaranteed upper bound, not an independence statement or a proof of minimality. At roots of unity, one can now ask which symmetries or distribution projections cancel these residual jets and leave only depth-one Hurwitz-zeta, Gamma, and generalized-Stieltjes data. Such a cancellation problem is distinct from the finite mixed-sign closure proved here.

7 Verification, source corrections, and further research

7.1 What the checks establish

The mathematical proofs above are independent of floating-point agreement. The supplied programs exercise three different layers of the argument.

Exact finite algebra. The cubic symbolic script treats $\gamma, L, \zeta(2), \zeta(3)$, the needed zeta derivatives, and Ω_3, J_3 as formal symbols. It reconstructs the four quadratic coefficients of H from the four stated restrictions and reduces the residual in (4.5) to the zero rational function. Its curve checks compare direct truncated quotient expansion with (5.3); they retain all path coefficients. The mixed-order script independently constructs the original coefficient convolution, the shuffle/partial-fraction formula, and the output after eliminating nonpositive trailing orders. Its 624 exact rational checks include coincident colors and repeated poles. A separate local-coefficient script checks the displayed Mellin principal parts and Gamma jets and the finite harmonic summation-by-parts algebra in Section 2. In total, the three exact suites record 760 passed checks: 16 cubic/ray identities, 624 mixed-order checks, and 120 local-coefficient checks.

A certified scalar evaluation. The code `verify_cubic_harmonic_bridge.py` combines Arb ball arithmetic with (2.10), using $N = 50$, $K = 24$ and 100 decimal digits of working precision. The stored enclosure is

$$Q \in 1.9662627343915343406643753384116314238258689764916868267443076 \pm 4.86 \cdot 10^{-62}.$$

The analytic tail contribution alone is below $1.26 \cdot 10^{-62}$. The reported enclosure includes rounding of its displayed center. The corresponding enclosures for $\eta(1)$ and Ω_3 are obtained using the proved exact identities. They are not independent proofs of those identities.

Independent numerical diagnostics. Ordinary high-precision quadrature of the regularized digamma cube and the harmonic Mellin integral agree through approximately 70 decimal places. A different evaluator computes $\mathcal{T}$ by integrating Jonquière’s series on $(0, 1)$ and an incomplete-Gamma tail on $(1, \infty)$. Symmetric differences with polynomial extrapolation check the third derivatives on six additional rays, including $(1, -2, 3)$. Their largest recorded absolute discrepancy is below $5.7 \cdot 10^{-37}$. An independent computation of I_2 gives the same second cubic coordinate through more than 30 decimal places. These latter calculations use non-interval arithmetic and truncated series; their digits are diagnostics, not certified error estimates. The split at one is strictly inside the Jonquière radius 2π .

The package records exact program versions, numerical parameters, machine-readable outputs, and build instructions. No finite test suite certifies the analytic convergence arguments: those arguments are supplied in the text. The exact symbolic checks certify the specified finite algebra, rather than all possible extensions of a theorem.

7.2 A source-status ledger for integration

The source snapshot is pinned by its full commit identifier, with blob hashes in `provenance/repository_inventory.json`. All 80 manuscript chapter files and all 11 incoming archives matched their recorded Git blob hashes. This establishes the identity of the source corpus; it is not a claim that every proof in that corpus has received a new exhaustive review. The mathematical audit concentrated on the harmonic normalization, the cubic completion, the local Tornheim decomposition, and the rational/colored reduction machinery used by the present results.

Existing target	Result in this report	Status to record
Coincident report, Q1: third diagonal derivative	Theorem 3.2 and the absolutely convergent harmonic series (2.3).	Reduced to $\eta(1)$; finite ordinary-constant closure remains open.
Coincident report, Q2: asymmetric cubic directions	Theorem 4.1, with two fixed coordinates and explicit coefficients; κ also has the convergent integral (4.19).	Complete spanning formula proved; arithmetic minimality unproved.
Coincident report, Q11: mixed integer inner signs Strict/inclusive harmonic conventions	Theorem 6.1, including arbitrary integer weights, coincident colors, repeated poles and all outer-order derivatives. (1.4) is retained exactly, including the sign and the $\zeta'(2, a)$ shift.	Finite closure proved, with a guaranteed depth bound. Existing normalization confirmed.
Gaussian S_6 and revised S_8 candidates	No exact target-vector certificate is supplied here.	Remain open; no theorem label should be promoted.

7.3 Corrections and qualifications worth preserving

No new false theorem was found in the specific repository results used here. Several distinctions are nevertheless necessary for a correct integration.

1. **Keep the full resonant numerators.** The cubic completion (3.3) subtracts $3h(t)^2/t$ and adds $3(1-t)\zeta(2-t)/(2t)$. Replacing their numerators by their constant terms changes

the answer. This is an existing warning in the coincident report, rederived here to protect the new harmonic bridge from a finite normalization error.

2. **Use the mixed pole rule only in its stated domain.** The rational sector has a finite simple-pole description. Its extension with positive uncolored inner orders can have higher pole order and infinitely many nonpositive poles. The explicit $Z_{1,1,0}(C;1)$ example proves this difference. The earlier rational theorem is not wrong; extending its pole statement without the logarithmic correction would be wrong.
3. **Separate coordinate rank from arithmetic independence.** Two formal quadratic coordinates of H survive the exact slice constraints, with slope coefficients abc and $c\Delta$. The theorem determines that reconstruction problem. It does not show that $\eta(1)$ or κ lies outside a chosen field of ordinary constants, or that they are independent of each other.
4. **Restrict before differentiating.** An admissible Tornheim ray is holomorphic at zero, while the full three-variable function is meromorphic there. The values $1/3$ on the diagonal and $5/12$ on the C axis are consistent. A curve tangent to $A + C = 0$ or $B + C = 0$ is outside Theorem 5.1; its value is not assigned by cancellation of a zero denominator in the displayed ray formula.

The incoming report already records corrections to the inspected Borwein–Dilcher author PDF [2, 6]. They should retain that attribution. In the integer Jonquière formula, the convention is $H_0 = 0$, and the coefficient of $x^{p-1} \log x$ is

$$B_p = \frac{(-1)^p}{\Gamma(p)}, \quad p \geq 1,$$

without an additional factor H_{p-1} . This follows at once from (6.21); $p = 3$ gives $-1/2$. For the ordinary convergent local power series, the splitting point must satisfy $0 < \theta < 2\pi$. These are previously documented corrections to the inspected author version, not newly discovered errors or assertions about every published version.

7.4 Proposed further research questions

The following problems are specific next steps. They are questions, not conjectured equalities unsupported by calculation.

1. Determine the arithmetic content of the harmonic bridge. Fix a constant algebra before attempting reduction, for example one generated over $\mathbb{Q}$ by ordinary Stieltjes constants, ordinary zeta derivatives at integers, logarithms and specified Barnes values. Does $\eta(1)$ have a finite expression in that algebra? The bridge gives three analytic realizations of the same coordinate: an ordered double-zeta Laurent coefficient, a convergent logarithmic harmonic series, and a normalized digamma-cube integral. Seek a functional identity relating two realizations without reintroducing the same coefficient under another name.

2. Relate the second cubic coordinate to harmonic Laurent data. The constant κ is a fixed asymmetric double-zeta derivative and also the convergent moment (4.19) plus classical terms. Can an ordered Hurwitz functional equation carry that derivative from the origin to a harmonic Stieltjes coefficient at $(1,1)$? A successful identity should state the exact finite correction and the direction of continuation. The question of a relation between κ and $\eta(1)$ is distinct from the proved two-coordinate spanning formula.

3. Classify useful cancellations among cubic directions. The conic $\Delta = 0$ removes κ from a single ray. More generally, a finite combination $\sum_j r_j W_{v_j}'''(0)$ has no displayed nonclassical

coordinate whenever $\sum_j r_j a_j b_j c_j = 0$ and $\sum_j r_j c_j \Delta_j = 0$. Classify short rational configurations with additional permutation, distribution, or character symmetry. This is a finite exact linear problem and can produce concrete new single-zeta identities without an independence assumption.

4. Construct the complete quartic directional layer. At fourth order the cubic homogeneous part of H enters. Determine its exact dimension after symmetry and the axis/stuffle restrictions, then choose independent anchor slices and derive the analogue of (4.5). The first goal is a proved finite coordinate formula with convergent integral representatives. A finite ordinary-zeta evaluation should be a separate theorem if achieved.

5. Extend the cubic bridge to a common Hurwitz shift. For $a > 0$, the coefficient $\eta(a)$ has exact shift and derivative relations in the ordered report. Apply the partial-fraction method to shifted poles and compare with the primitive $\mathcal{P}_3(a+1) - \mathcal{P}_3(a)$. Determine the explicit boundary terms and whether products of shifted digamma and polygamma functions give a useful identity involving $\eta(a)$. The unit-coordinate finite part at a singular endpoint and an ordinary integral over a positive shifted interval must stay distinct.

6. Obtain a general power-to-harmonic formula. For each $k \geq 4$, the principal parts of $\psi(z)^k$ are finite generalized-harmonic polynomials. The two-point contour argument still suggests a normally convergent subtraction. Carry out the finite summation-by-parts reduction explicitly for $k = 4$ and identify the required harmonic Laurent coefficients or higher-depth data. The target is a compact identity with every constant fixed, rather than an unspecified Mittag-Leffler entire remainder.

7. Find depth-reducing projections in the mixed sector. At root-of-unity colors, the finite reduction outputs a finite list of nested functions and their first-order derivatives. Construct exact matrices for character projections and distribution identities on that list. Their kernels can identify combinations whose depth drops to one, where Hurwitz-zeta, Gamma and generalized-Stieltjes formulas become available. Any claimed minimal depth must specify the relation space being used.

8. Evaluate selected holomorphic Laurent coefficients. Theorem 6.4 gives all principal parts but deliberately does not promise finite depth-one formulas for the regular terms. Begin with $Z_{1,1,0}(C; 1)$ at $C = 1, 0, -1$: derive explicit convergent kernels for its constant and linear Laurent coefficients and relate them to depth-three harmonic Stieltjes conventions. This separates the already solved local pole algebra from new global arithmetic.

9. Normalize ordinary primitives under reflection and multiplication. The primitive $\mathcal{P}_3$ fixes its integration constant at the singular origin, while (6.30) fixes a holomorphic Euler primitive at zero. Determine exact reflection and multiplication relations for these normalizations, including their polynomial and logarithmic correction terms. At low orders, compare with Barnes multiple-Gamma primitives only after matching the additive constants and branches.

10. Resolve tangential spectral paths. The transverse formula requires $(a+c)(b+c) \neq 0$. For a curve whose first nonzero contact with one polar divisor occurs at order $r > 1$, determine when the numerator cancels enough terms to leave a removable restriction, and which path jets fix the resulting finite value. This is a divisibility and coefficient-extraction problem, not an instruction to apply the transverse formula at a zero denominator.

11. Return to the exact Gaussian targets. The frozen S_6 and revised S_8 vectors still need exact certificates in proved relation spaces. Use the current reduction and projection machinery to propose additional rows, proving the regularization prescription for each row. A numerical integer relation is a candidate; a successful proof must combine exact rows to the exact target. An unsuccessful restricted search should report its precise space and a separating functional, not an unrestricted impossibility claim.

12. Formalize the finite proof interfaces. The partial-fraction extraction, shuffle multiplicities, finite antidifferences, Gamma jets, and cubic coefficient elimination are all terminating algebra. Separate them from analytic lemmas for normal convergence, Mellin continuation and endpoint constant terms. A formal implementation can then check the finite identities while explicitly declaring the analytic hypotheses on which their applications depend.

These questions preserve the emphasis on exact identities and normalizations. Error bounds appear in this report only to certify the numerical evaluation of an already proved identity.

References

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